MASKAGLIA

A New, Efficient Approach to Masked Discrete Gaussian Sampling

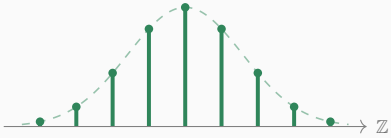
Calvin Abou Haidar[1] Thomas Espitau[2] Clément Hoffmann[1] Mehdi Tibouchi[1]

CRYPTO 2026 — Santa Barbara

[1]NTT Social Informatics Laboratories [2]PQShield

Discrete Gaussians in cryptography

ESSENTIAL

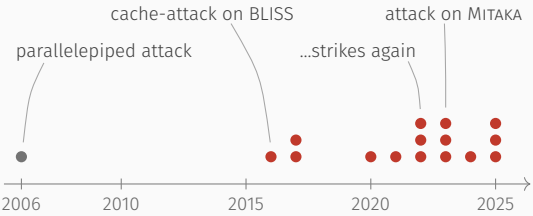


conservative parameters (FRODOKEM)

essential for security (HAWK, FALCON)

advanced applications (FHE, IBE)

HISTORICAL WEAK POINT



Dilithium & Raccoon: designed to *avoid* Gaussians

constant-time sampling: settled

leakage: masking is the open problem

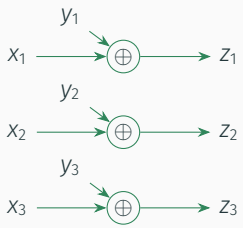
Masking in one minute

$[\![x]\!] = (x_1, \ldots, x_d), \quad x = x_1 \oplus x_2 \oplus \cdots \oplus x_d$ any $d-1$ shares learn nothing

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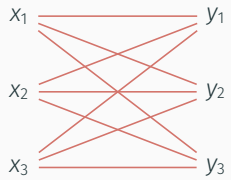
$x \oplus y = z$



linear (XOR, shifts, ...): share-wise

\+ no fresh randomness

$x \wedge y$



d^2 partial products

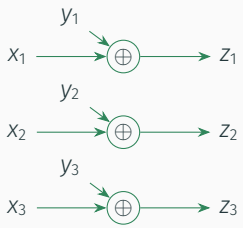
non-linear (AND): all cross pairs

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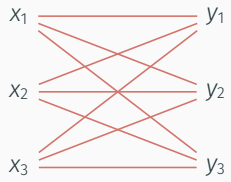
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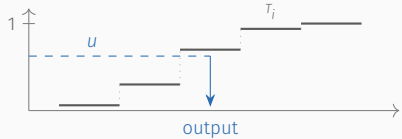
\+ fresh randomness

For this paper

Compute the sampler as a Boolean circuit with **as few SecAnds as possible**.

- we use bitslicing: arbitrary depth + amortized cost ($\div 32$)
- t-probing security proofs rely on the composability of PINI gadgets

State of the art: CDT-based sampling



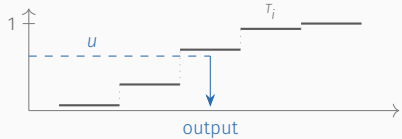
Inverse sampling.

$T_i = \Pr[X \leq i]$: CDT values

draw u uniform,

return the first i with $u \leq T_i$

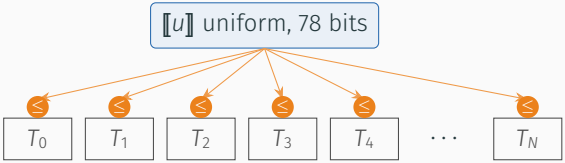
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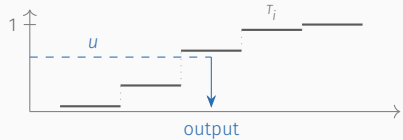
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linear scan [Gérard–Rossi '19]
$\Rightarrow N$ masked comparisons

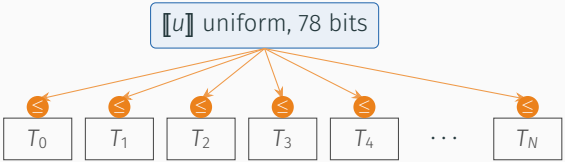
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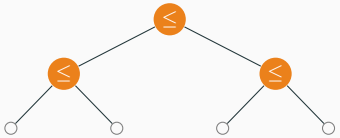
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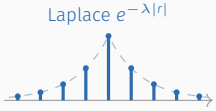


binary search [Eid et al. '26]
$\Rightarrow \lceil \log N \rceil$ comparisons

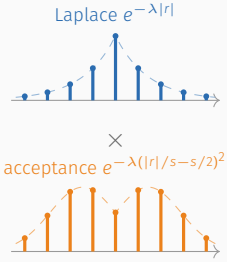
pivots must be selected *masked* too;

cost $\approx$ #comparisons $\times$ cost(78-bit $\leq$ $\approx$ 77 SecAnd)

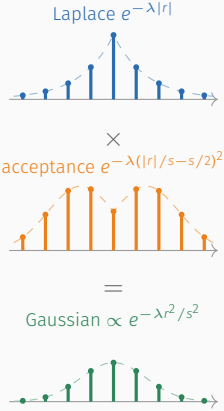
A rejection-based sampler



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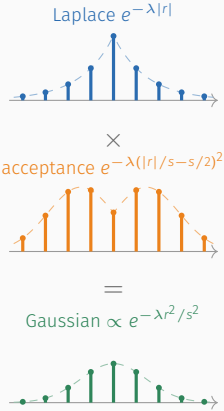
A rejection-based sampler



$$\underbrace{e^{-\lambda|r|}}_{\text{propose}} \cdot \underbrace{e^{-\lambda(|r|/s-s/2)^2}}_{\text{accept}} \propto \underbrace{e^{-\lambda r^2/s^2}}_{\text{obtain}}$$

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1 **repeat**
2 $\quad Y \xleftarrow{\$} \mathrm{Lap}_c(\lambda)$
3 $\quad X \xleftarrow{\$} \mathrm{Exp}(\lambda)$
4 **until** $X \geq \left(|Y - c|/s - s/2\right)^2$
5 **return** Y $\qquad \sim \mathcal{N}_{c,\sigma}$ on $\mathbb{R}$, $D_{\mathbb{Z},c,\sigma}$ on $\mathbb{Z}$

the continuous version is an old sampler [Butcher '61]

$\lambda = \ln 2$: (almost) everything becomes bits

the identity holds for *any* λ, so pick $\lambda = \ln 2$: $e^{-x \ln 2} = 2^{-x}$, everything becomes binary

$X = -\log_2 V \overset{\$}{\leftarrow} \mathrm{Exp}(\ln 2)$, $V \overset{\$}{\leftarrow} \mathcal{U}(0,1)$: $X =$ $K = \lfloor X \rfloor$ $\mathrm{Geom}(1/2)$ $+$ $U = \{X\}$ transcendental function

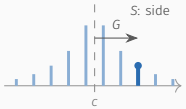
easy to mask | maskable for favorable parameters | the hard part

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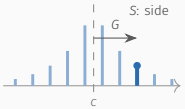
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U is the **only** part hard to mask

An integer test

$s, c \in \mathbb{Q}$: $\left(|Y - c|/s - s/2\right)^2 \overset{\text{notation}}{=} L_Y^2 \,/\, K_0$ K_0 fixed, L_Y affine in $|Y-c|$ are integers, precomputed

$$\text{reject} \iff X < \left(|Y - c|/s - s/2\right)^2 \iff K_0\left(K + U\right) < L_Y^2$$

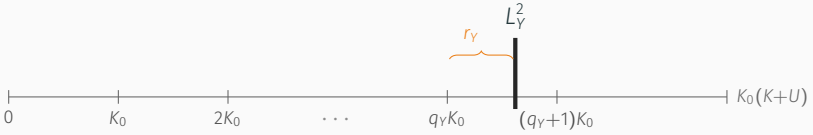
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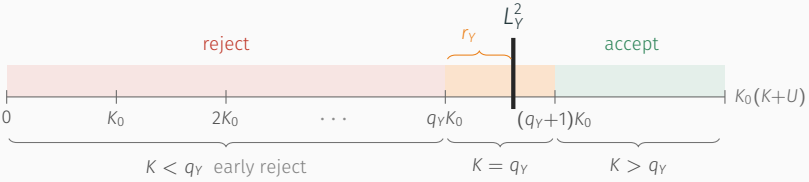


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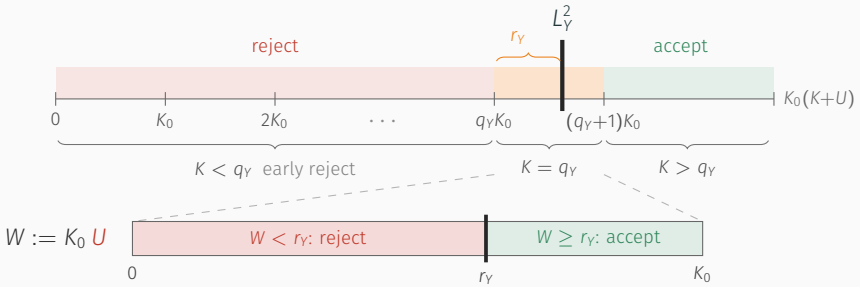


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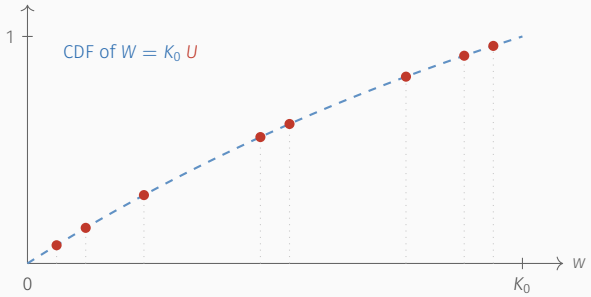
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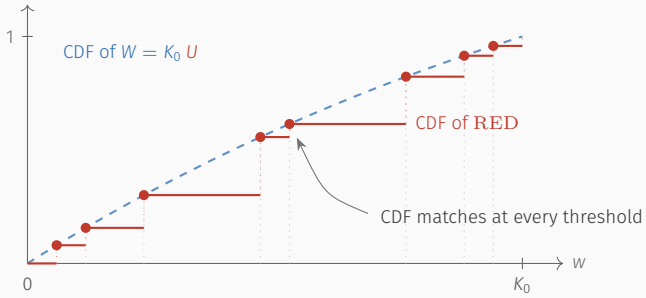
all that remains of U: **one integer comparison**, $W < r_Y$

The Rejection-Equivalent Distribution (RED)



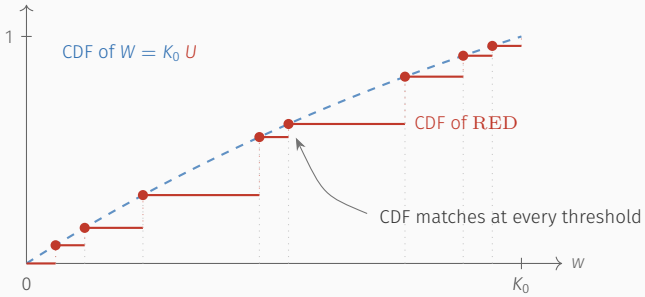
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The Rejection-Equivalent Distribution (RED)



U — real
transcendental

swap *exactly*

RED — discrete
same answers, always

sampling RED: **inverse sampling**
same as a CDT, but with

fewer values and smaller integers

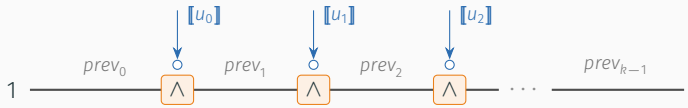
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we now sample *exactly* the target distribution with only integers

finite precision is also handled: the analysis combines Rényi divergence and algebraic arguments

Masked geometric sampling: scan for the first 1

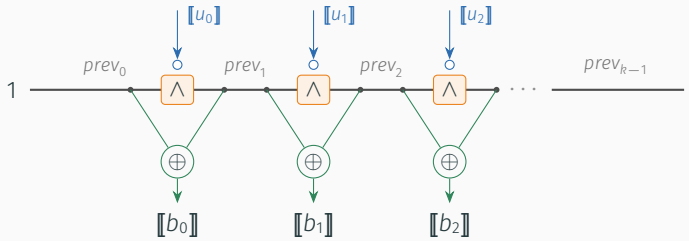
$\mathrm{Geom}(1/2)$: the position of the first 1 in a fresh uniform bitstream



$prev_{i+1} = prev_i \wedge \neg\, u_i$
each ∧: 1 SecAnd
$k-1$ total (the first is a free ¬)

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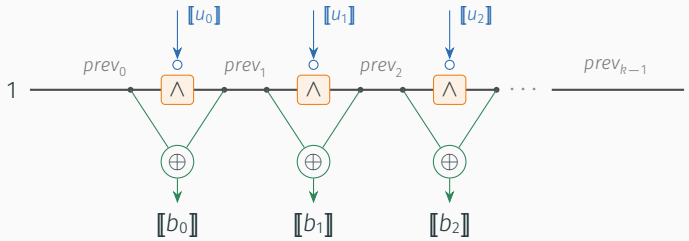
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$b_i = prev_i \oplus prev_{i+1}$
linear

exactly one b_i equals 1: the one at position G

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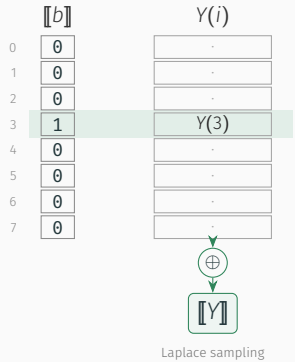
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output: $i \cdot ⟦b_i⟧$ multiplying by a public constant is free

Free lookups in precomputed tables

$$[\![f(G)]\!] = \bigoplus_i f(i) \cdot [\![b_i]\!]$$ public constant × masked bit: share-wise = free

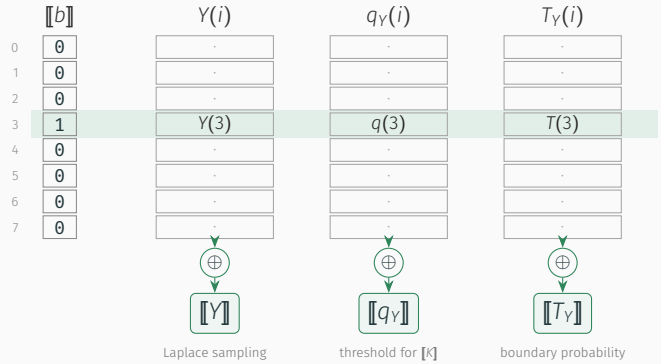


precomputed, public

the only surviving row

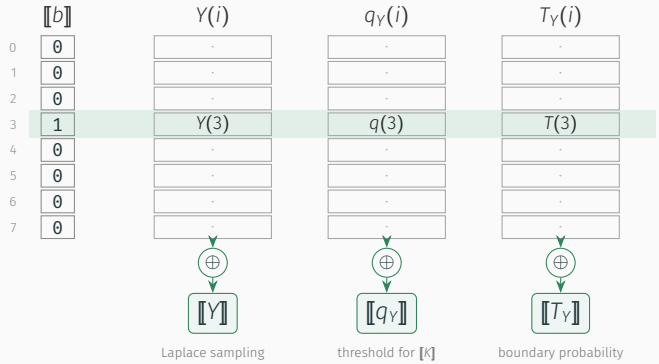
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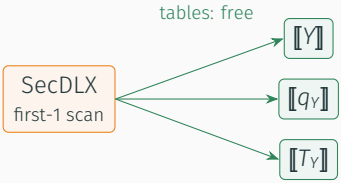


all the heavy arithmetic lives in **public tables** ⇒ 0 extra SecAnd

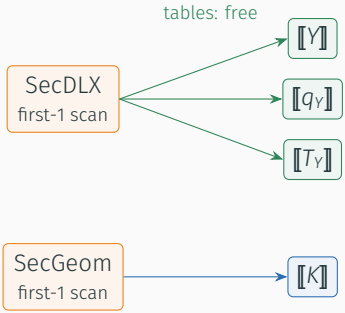
Geom gives one side of the Laplace only: to handle the side S, tables are doubled.

With symmetry (e.g. Hawk, $c = \frac{1}{2}$) this is totally free.

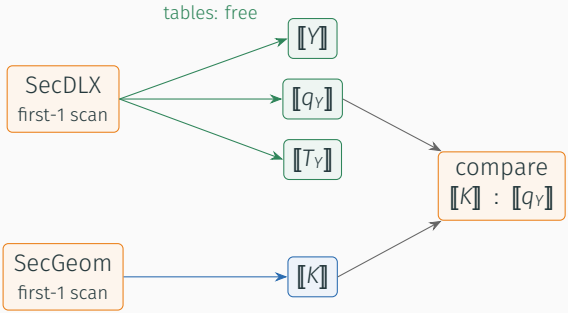
The MASKAGLIA gadget



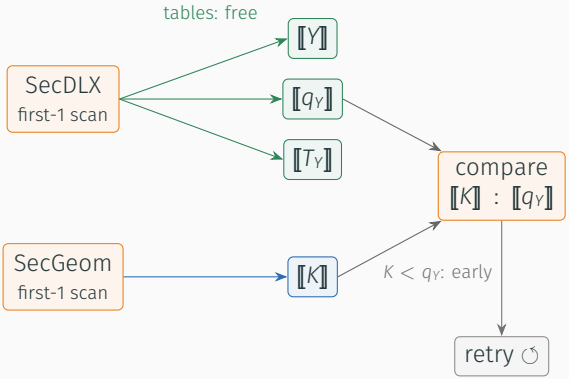
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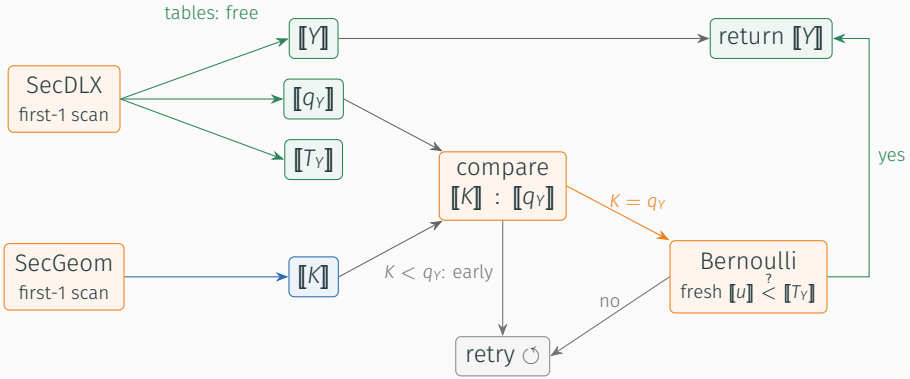


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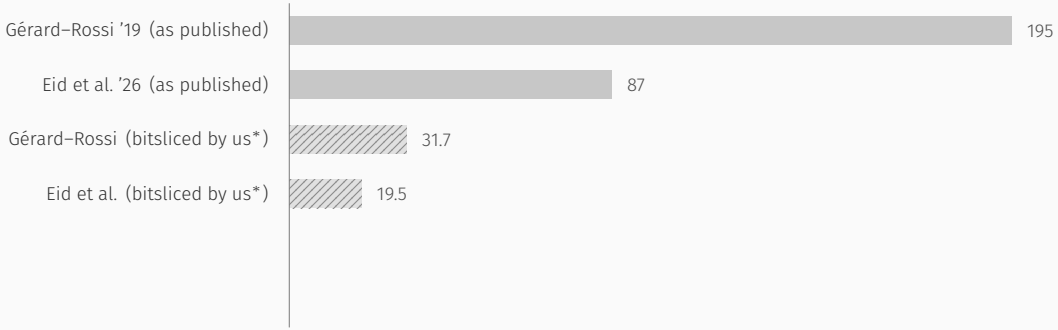
SecUnmask: # iterations is public

The Maskaglia gadget



three functions of a geometric, another (smaller) geometric, one comparison, one Bernoulli

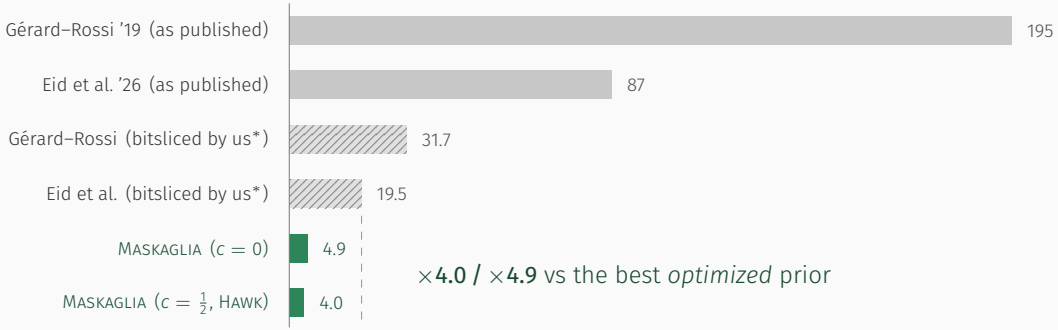
Hawk: < 5 amortized SecAnds per sample



amortized SecAnd_{32} per sample: gate counts, independent of the masking order

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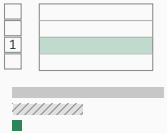


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Conclusion

a non-CDT paradigm: Laplace × quadratic rejection, discretized via many tricks



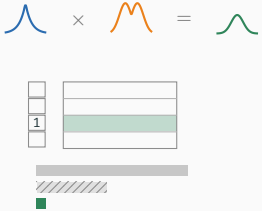
lookups in precomputed tables: public post-processing is free under masking

HAWK: ×4–5 fewer SecAnds per sample than *optimized* prior work

extra applications:

- wide σ: by convolution of narrow samples
- FALCON: the base sampler works as is; secret parameters remain the hard part
- FRODOKEM: excluded by the FO transform (same randomness usage required)

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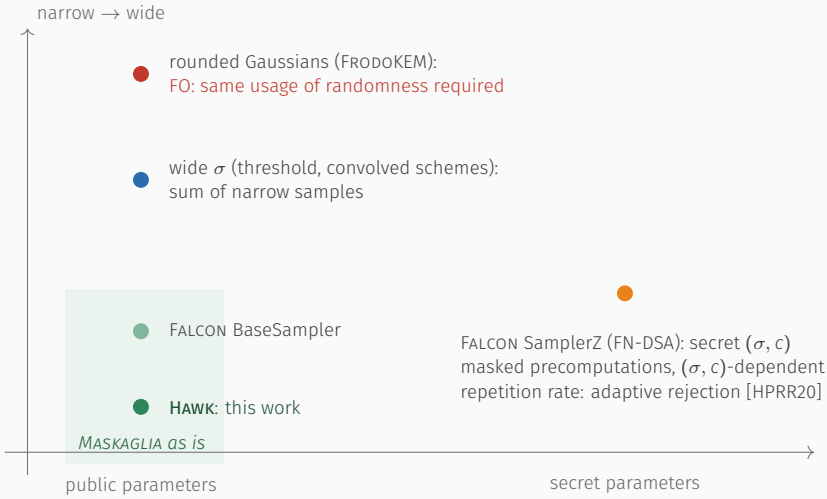
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Thank you

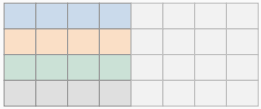
Backup — Scope: the full map



Backup — Bitslicing: why costs divide by w

our circuits are made of **1-bit gates** — pack w independent samples per machine word

one register per *value*

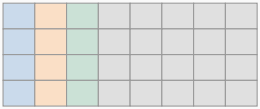


4/32 bits busy, padding idle

transpose

amortized

one register per *bit plane*



bit 0 of every sample
bit 1 of every sample
bit 2 of every sample
bit 3 of every sample

all $w = 32$ lanes busy

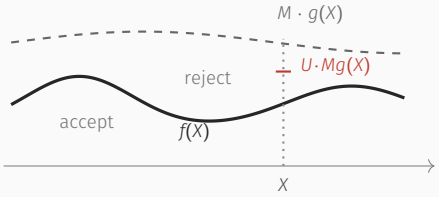
one SecAnd_1^d on bit planes $= 32$ samples at once $\Rightarrow$ every gate count $\div w$

requires high parallelism, which lattice-based schemes provide (hundreds of samples per signature);

the transposition overhead is word-level and amortized across the batch

Backup — Rejection sampling

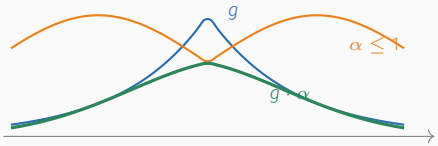
with an **envelope** $M \cdot g$



$X \xleftarrow{\$} g$, $U \xleftarrow{\$} \mathcal{U}(0,1)$: accept iff $UMg(X) \leq f(X)$

output $\sim f$ — $\mathbb{E}[\text{trials}] = M$

with an **arbitrary** acceptance α



$X \xleftarrow{\$} g$: accept with probability $\alpha(X)$

output $\propto g \cdot \alpha$ — $\mathbb{E}[\text{trials}] = 1/\int g\alpha$

MASKAGLIA: $g =$ Laplace, $\alpha =$ quadratic rejection condition $\Rightarrow g\alpha =$ Gaussian, exactly

Backup — SecDLX: the real pseudo-code

Input: public pairs $(f_{0,j}, f_{1,j})_{j<n}$, center $c \in \mathbb{Q}$ — Output: $[\![f_{b,j}(p)]\!]_{j<n}$, $p \sim$ trunc. $\mathrm{Geom}(\frac{1}{2})$ on $\{0..k{-}1\}$, b secret side

$[\![u]\!] \leftarrow \mathrm{SecUnif}^d(k)$

$[\![S]\!] \leftarrow \mathrm{SecBernoulli}^d(p_\delta),\ \delta = c - \lfloor c \rfloor$ — secret side of the center

$[\![prev]\!] \leftarrow \neg\, [\![u_0]\!]$ — free

$[\![\sigma]\!] \leftarrow \mathrm{SecAnd}^d([\![u_0]\!], [\![S]\!])$ — branch mux

for $j < n$: $[\![r_j]\!] \leftarrow \mathrm{Select}(f_{0,j}(0), [\![u_0]\!] \oplus [\![\sigma]\!]) \oplus \mathrm{Select}(f_{1,j}(0), [\![\sigma]\!])$ — public × bit: linear

for $i = 1 \ldots k-1$:

  $[\![b]\!] \leftarrow \mathrm{SecAnd}^d([\![prev]\!], [\![u_i]\!])$ — first-1 indicator

  $[\![\sigma]\!] \leftarrow \mathrm{SecAnd}^d([\![b]\!], [\![S]\!])$ — 1 AND → both branches

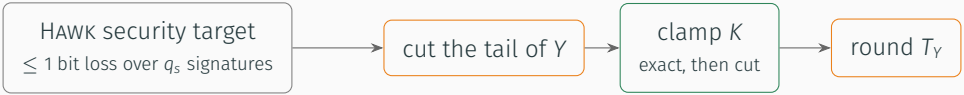
  for $j < n$:

    $[\![r_j]\!] \oplus\!= \mathrm{Select}(f_{0,j}(i), [\![b]\!] \oplus [\![\sigma]\!]) \oplus \mathrm{Select}(f_{1,j}(i), [\![\sigma]\!])$ — $[\![b]\!] \oplus [\![\sigma]\!] = [\![b \wedge \neg S]\!]$

  $[\![prev]\!] \leftarrow [\![prev]\!] \oplus [\![b]\!]$ — flag update: linear

return $([\![r_0]\!], \ldots, [\![r_{n-1}]\!])$

Backup — Tailcuts: the Rényi workflow



every lossy step certified in Rényi divergence

we provide a script: the distribution in, all precision parameters out